

Official System Architecture & Operational Blueprint

Project Atlas v1.0

Autonomous Multi-Tenant Restaurant Operating System & Autonomous Intelligence Engine. Complete End-to-End Workflow Guide for Platform Operators, Restaurant Owners, Kitchen Chefs, Floor Waiters, and Cashiers.

v1.0

PRODUCTION
LIVE

ARCHITECTURE

NestJS + Next.js 16

DATA LAYER

PostgreSQL + Prisma ORM

MULTI-TENANCY

Strict Row-Level Isolation

INTELLIGENCE

Groq / OpenAI Context
Copilot

01 System Hierarchy & Data Isolation

Project Atlas enforces a hierarchical multi-tenant structure designed for independent restaurant scalability, multi-branch expansion, and total data isolation across tenants.

PLATFORM ADMIN (Global Infrastructure, Global Monitoring, Support Desk, Subscription Tiers) — **TENANT** (Billing Organization, Subscription Account, Master Owner) — **RESTAURANT** (Concept Branding, Master Menu, AI Copilot, Automation Rules) — **BRANCH** (Physical Location, Local Inventory, Operating Hours) — **DINING AREA** (Floor Plan Section: Indoor, Rooftop, Courtyard) — **TABLE** (Unique QR Session Token `tbl\_xxxx`, Real-time Cart) — **ORDERS & PAYMENTS** (KDS Pipeline: PENDING → PREPARING → READY → SERVED → COMPLETED)

Multi-Tenant Guards

Every API route enforces TenantAccessGuard, RestaurantAccessGuard, and BranchAccessGuard. Cross-tenant leakage results in immediate HTTP 403 Forbidden.

Session QR Engine

Tables have encrypted public tokens (e.g., tbl_33c549...). Customers order directly without requiring mobile app downloads or logins.

Autonomous AI Brain

Integrated conversational copilot and scheduled cron automations actively analyze revenue, low stock, and peak sales hours.

02 Platform Admin Panel Flow (`/platform-admin`)

The Platform Admin Desk provides global operators with centralized observability, SaaS subscription enforcement, customer support resolution, and infrastructure health monitoring.

1

Global Telemetry & Platform Overview

PLATFORM_ADMIN

Real-time aggregate across all tenants: total onboarded restaurants, monthly recurring revenue (MRR), total active tables, order throughput per minute, and live P50/P95 latency percentiles.

GET /v1/dashboard/platform-overview

Active Tenants

Platform Sales (₹)

Health Probes

2

Tenant & Restaurant Lifecycle Management

PLATFORM_ADMIN

Inspect tenant accounts, audit restaurant configurations, manage subscription plan tiers (Free, Growth, Enterprise), adjust seat quotas, and activate feature flags (e.g. AI Copilot, Multi-Branch, Webhook integrations).

GET /v1/tenants

PATCH /v1/subscriptions/:id

Plan Quotas

Feature Toggles

3

Customer Support & Incident Resolution Desk

PLATFORM_ADMIN

Central queue of incident tickets submitted by restaurant owners. Platform engineers filter by category (TECHNICAL, BILLING, HARDWARE), prioritize by SLA urgency (URGENT, HIGH), and submit resolution notes.

GET /v1/support/admin/tickets

PATCH /v1/support/tickets/:id/resolve

SLA P0-P3 Resolution

4

Observability, Database Backups & Infrastructure

PLATFORM_ADMIN

Live monitoring of PostgreSQL connection pools, Redis queue latency, and automated disaster recovery backups (pnpm db:backup) with foreign key consistency verification.

GET /v1/health/live

GET /v1/health/ready

Disaster Recovery Snapshots

The core workflow for restaurant owners from initial onboarding to day-to-day operations, menu engineering, and AI-driven growth.



6-Step Guided Onboarding (`/onboarding`)

1. Profile & Concept (Identity, Currency INR ₹)
2. Floor Layout & Dining Areas (Courtyard, Rooftop)
3. Dynamic QR Generation (Table 01 to T-50)
4. Menu Catalog & Signature Dishes
5. Staff Invitation (Kitchen, Waiter, Cashier)
6. Payment Gateway & Go-Live Activation



Real-Time Operations Command (`/dashboard`)

- Live Sales Revenue (₹) & Order Count
- Active Occupied Tables & Table Turnover
- Hourly Sales Velocity Charts (Peak hours)
- Best-Selling Menu Items & Category Contribution
- Real-time synchronization with KDS and POS



Menu Engineering & Catalog (`/menus`)

- Multi-category master catalogs with nested item options
- Price modifiers, dietary flags (Veg, Non-Veg, Vegan)
- Real-time stock status (In Stock / 86'd Out of Stock)
- Instant live sync to table QR ordering



AI Copilot & Automation Engine

- Natural language queries: "*Which dishes drove the most profit yesterday?*"
- Scheduled crons: Nightly 11 PM Revenue Report
- Stock depletion alerts triggered automatically
- AI Operational recommendations generated every Monday

Guest QR Ordering Lifecycle (Zero App Download)

The friction-free customer ordering experience designed for mobile browsers with zero app installation.

1

Step 1: Scan Table QR & Session Handshake

GUEST / DINER

Guest points phone camera at table sticker. The encrypted token URL (`/t/tb1_4ec99d...`) initiates a session with the restaurant's active branch and table metadata.

`GET /v1/public/tables/:token/session`

Encrypted Token

Table Identification

2

Step 2: Digital Menu Exploration & Customization

GUEST / DINER

Guest browses high-res dishes, filters by dietary preferences, selects portion sizes, and adds special chef notes (e.g. *"Extra spicy, no cilantro"*).

`GET /v1/public/tables/:token/menu`

Dietary Badges

Real-Time Pricing

3

Step 3: Shared Table Cart & Order Placement

GUEST / DINER

Items are committed to the synchronized table cart. Guest reviews subtotal, applicable taxes (CGST/SGST), and taps **"Submit Order"** to generate order `#AT-XXXXXX`.

`POST /v1/public/tables/:token/cart/items``POST /v1/public/tables/:token/orders`

Instant Order Ticket

4

Step 4: Live Order Tracking & Service Calls

GUEST / DINER

Live status bar updates: `PENDING` → `PREPARING` → `READY`. Guest can request table service (*"Call Waiter"*, *"Water Refill"*, *"Request Bill"*) at any moment.

`GET /v1/public/tables/:token/orders/:id``POST /v1/public/tables/:token/call-waiter`

05 Floor & Kitchen Operations Workflow

Real-time synchronization across Kitchen Display Systems (KDS), Waitstaff floor terminals, and Cashier POS settlement counters.

STAGE	INTERFACE / ROUTE	ACTOR	OPERATIONAL ACTION & STATE TRANSITION
1. New Order Ticket	/kitchen (KDS)	Kitchen Expeditor / Chef	Audio chime triggers. Ticket arrives under PENDING with table number, quantities, and dish notes.
2. Cooking in Progress	/kitchen (KDS)	Line Chef	Chef clicks " Start Cooking ". Ticket moves to PREPARING . Prep timer begins tracking kitchen efficiency.
3. Food Plated & Ready	/kitchen (KDS)	Line Chef	Chef clicks " Ready for Pickup ". Ticket transitions to READY . Waitstaff terminal receives visual notification.
4. Table Delivery	/waiter (Floor App)	Floor Waiter	Waiter picks up hot dishes, delivers to table, and marks ticket SERVED .
5. Bill & Settlement	/cashier (POS)	Cashier / Manager	Cashier reviews table total, applies discounts, processes Payment (Cash, UPI QR, Card POS), and marks order COMPLETED .

06 Role-Based Access Control (RBAC) Matrix

Project Atlas enforces strict granular permissions mapped to specific operational roles.

ROLE	PRIMARY WORKSPACES	MENU / PRICING	KDS DISPLAY	ORDERS & BILLING	AI COPILOT	PLATFORM ADMIN
PLATFORM_ADMIN	/platform-admin, /docs	Global Audit	Global View	Global Telemetry	Full Access	FULL CONTROL
OWNER / TENANT_ADMIN	/dashboard, /onboarding, /menus	CREATE / EDIT	FULL ACCESS	FULL ACCESS	FULL ACCESS	NO ACCESS
MANAGER	/dashboard, /orders, /inventory	EDIT ITEMS	FULL ACCESS	FULL ACCESS	READ-ONLY	NO ACCESS
KITCHEN	/kitchen (KDS)	Read Only	MANAGE KDS	Status Update	NO ACCESS	NO ACCESS
WAITER	/waiter, /tables	Read Only	Read Status	TAKE / DELIVER	NO ACCESS	NO ACCESS
CASHIER	/cashier, /orders	Read Only	Read Status	SETTLE BILLS	NO ACCESS	NO ACCESS
GUEST (Anonymous QR)	/t/[token], /t/[token]/cart	Read Live Menu	NO ACCESS	SUBMIT ORDER	NO ACCESS	NO ACCESS

07 Production Resilience & Disaster Recovery

Multi-Tenant Security Guarantees

- Zero Cross-Tenant Leakage:** Every query in Prisma includes where: { restaurantId, tenantId } automatically scoped by JWT context.
- Token Integrity:** Cryptographically signed JWT access tokens with rotating refresh cookies and instant token blacklisting.
- Rate Limiting & DDoS Shield:** Tiered Redis rate limiters protecting public QR endpoints (100 req/min) and internal APIs (1000 req/min).

Automated Relational Backups

Execute full snapshot drills using `pnpm db:backup`. Snapshots 13 core relational tables and runs automated foreign key referential integrity checks before saving to encrypted storage.

Zero-Downtime Rolling Restarts

Docker Compose healthchecks ensure new API containers are fully warmed and passing PostgreSQL / Redis readiness probes (`/health/ready`) before traffic is cut over.